

Optimal Feeder Routing in Distribution System Planning Using Dynamic Programming Technique and GIS Facilities

Nicholas G. Boulaxis, *Student Member, IEEE*, and Michael P. Papadopoulos, *Member, IEEE*

Abstract—Optimal feeder routing is an important part of the general optimal distribution network planning. This paper proposes a new algorithm for the optimal feeder routing problem using the dynamic programming technique and geographical information systems (GIS) facilities. All practical issues, such as cost parameters (investments, line losses, reliability) and technical constraints (voltage drop and thermal limits), as well as physical routing constraints (obstacles, high-cost passages, existing line sections) are taken into consideration. The algorithm developed is validated comparing its results for a simplified study case, with those obtained by an established solver. The effectiveness of the algorithm is further illustrated for a “real-world” study case.

Index Terms—Dynamic programming, geographic information systems (GIS), planning, power distribution planning, routing.

I. INTRODUCTION

THE OPTIMAL distribution system planning is recognized as a very complex problem because of the large number of variables and candidate network configurations, related to the characteristics of the geographical area and the consumer loads served. An extensive presentation of the problem is given in [1], while several different approaches using mathematical programming techniques have been published, [2]–[15]. In [2]–[4], the problem of substation sizing and location is addressed. In [5]–[15], the main feeder network is also considered. In the first publications on the subject, simple linearized cost functions and constraints were used and mixed integer programming, transshipment methods and branch and bound techniques were extensively applied. In subsequent work, more complicated formulations and methods were applied, using more advanced techniques.

Because of the complexity of the distribution networks and the large number of variables involved, the planning methods developed mainly address the problem of substation location/sizing and main feeder transfer capabilities and/or main roots. Besides, the possible feeder roots are predetermined and the optimum one is selected.

Recently, an extensive application of geographical information systems (GIS) has enabled the representation of distribution systems in more detail, for optimum operation and planning purposes. On the other hand, the analysis requirements have become more complex, because of the constantly increasing

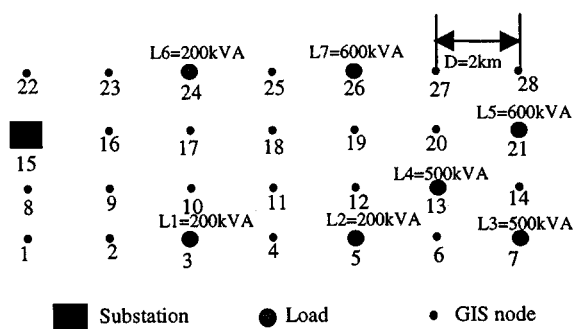


Fig. 1. An example of routing problem.

number of small, dispersed generation units, such as wind turbines, photovoltaics, etc.

In this paper, a method for the optimum planning, sizing and routing of a medium voltage (MV) radial network and the corresponding algorithm are presented. Emphasis is given on the optimum routing of the feeder, where different types of consumers or self-producers are connected. The optimum design of the network is achieved by minimizing the total cost of the radial feeders for a period of N years, over which the locations and characteristics of the loads are known. The method can be applied in any other type of network.

The main advantage of the developed algorithm over previously published approaches is that all cost parameters and technical or physical constraints are taken into consideration. In our knowledge, this has not been obtained until now, especially both voltage constraint and routing obstacles.

II. PRINCIPLES OF THE METHOD

First, the site, the year of installation, and the serving area of the high voltage/medium voltage (HV/MV) substation are determined, e.g., as in [14], [15] or otherwise. The objective is then to optimize the number, sizing, and routing of the feeders, subject to certain constraints. The optimal network configuration is obtained, when all the loads are connected, so that the minimum cost for the whole period of study is achieved.

In order to present the principles of the method, the elementary network of Fig. 1 is considered, where the locations of the HV/MV substation and the seven loads are indicated. Each node, numbered from 1 to 28, corresponds to a pixel of the GIS grid.

The proposed method of optimal feeder routing is based on dynamic programming. Using dynamic programming

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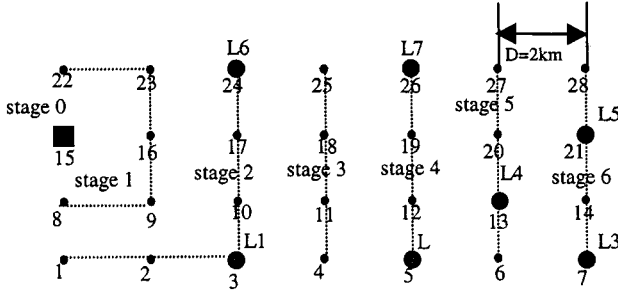


Fig. 2. Stages definition.

terminology [16], the following basic elements of the feeder model are defined:

- **Markovian Property (Principle of Optimality):** In dynamic programming, given a current state S_k , a policy for the remaining stages must be independent of the policy adopted in previous stages. In the case of a routing problem in a radial network, this prerequisite is satisfied by specifying that a node must feed specific loads. Hence, the modeling of the routing problem must be based on this principle, in order for the Markovian property to be valid.

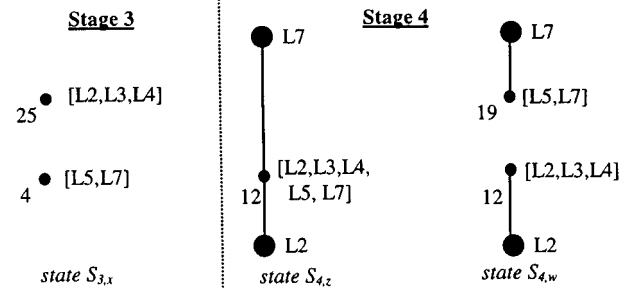
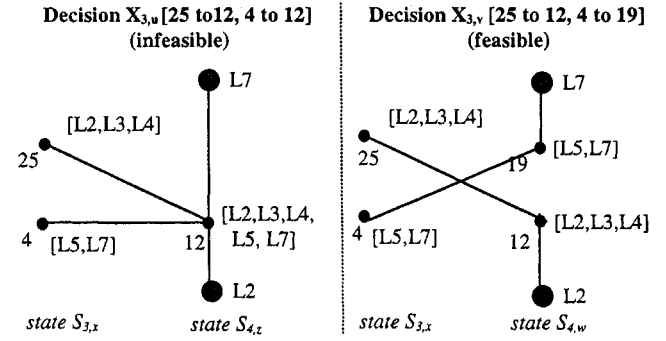
- **Stage k :** A stage is defined as a subset of the GIS network nodes. The GIS grid nodes are divided in a number of stages. The nodes of each stage, M_k , are perimetrically arranged around the substation. The distribution of the nodes in stages is generally arbitrary, but the following three rules must be followed.

- The stage enumeration starts from the HV/MV substation node to the remote loads. The first stage (zero) contains only the HV/MV substation node, while the last one (n) contains the geographically more distant from the substation load(s).
- Each node belongs to only one stage.
- Each node of the stage k is geographically closer to the HV/MV substation than any node of the stage $k + 1$.

Applying these rules, GIS nodes of Fig. 1 are divided into seven stages, indicated by dotted lines in Fig. 2.

- **State S_k :** The state S_k in a stage k is defined as a subset of the nodes of the stage, which “feed” all the loads of stages $k, k + 1, \dots, n$ in a specific way. Each subset includes M_{ki} nodes ($M_{ki} \leq M_k$). The number of subsets is equal to the possible combinations of M_k per 1, per 2, $\dots$, per M_k , equal to $2^{M_k} - 1$. In each combination, the subset of M_{ki} nodes feed the L_{kn} loads of stages k to n . The number of feeding arrangements is $M_{ki}^{L_{kn}}$. Each one of them is a state S_k . Hence, although all states S_k of stage k feed the same loads, the distribution of the loads between the nodes is different for each state. For example, the stage $k = 4$ includes the $M_4 = 4$ nodes 5, 12, 19, and 26, Fig. 2. The number of combinations of these nodes is $2^4 - 1 = 15$. A random combination i , e.g., $M_{4i} = 2$ (nodes 5 and 19) will produce $2^5 = 32$ states. In Fig. 3, two of them are presented, as well as another one, randomly selected, for stage $k = 3$.

“Feeding” the system loads by a node can be made directly or through other nodes of the same or other stages. However, only loads in the same stage k are fed directly from the stage nodes. It is also noted that within each state S_k , a “local” optimization is

Fig. 3. Two randomly selected states (z, w) of stage 4 and another one (x) of the stage 3.Fig. 4. The two decisions $X_{3,u}, X_{3,v}$ from state $S_{3,x}$ to states $S_{4,z}, S_{4,w}$.

made regarding the connection of the stage k loads to the nodes of state S_k .

- **Decision X_k :** In dynamic programming, it is defined as the selection of going from a state of stage k to another state in stage $k + 1$. In the routing problem, going from a state S_k to a state S_{k+1} is equivalent to connecting the nodes of these two states. Therefore, two nodes, which feed at least one load in common, must be connected to each other. In this way, the nodes of stage k also feed the loads fed by the nodes of stage $k + 1$. Using this principle, the interconnection of the nodes of two states is unique. In Fig. 4, two decisions $X_{3,u}$ and $X_{3,v}$ are presented, for going from state $S_{3,x}$ to states $S_{4,z}$, $S_{4,w}$, respectively. Given that more than one nodes of stage k can feed the same node of stage $k + 1$, the network radiality constraint can be violated. Consequently, a decision is characterized as “feasible,” when each node of stage $k + 1$ is connected to only one node of stage k .

- **Objective Function, Z :** The cost of a decision X_k , i.e., the cost of going from a state S_k to a state S_{k+1} in the next stage, is calculated by the “objective function” $Z_k(S_k, X_k)$, defined as follows:

$$Z_k(S_k, X_k) = ([\text{Connection cost}] + [\text{Power transmission cost}] + [\text{Reliability cost}] + [\text{Constraint cost}]) \text{ of } (S_k, X_k) \quad (1)$$

where

Connection cost is the construction cost of the lines required to interconnect the nodes of states S_k and S_{k+1} , as

well as the cost for the connection of the loads of stage k , for the whole study period. Already existing line sections are taken into account.

Power transmission cost includes the cost of the power losses associated with the decision X_k and the connection cost of the stage k loads for the period studied.

Reliability cost is the cost of nondistributed energy, due to line or other faults, related with decision X_k , and the connection of the loads of stage k .

Constraint cost is a virtual cost introduced in order to exclude decisions, which violate operational or other constraints. More specifically, the following constraints and corresponding virtual cost functions are defined:

“Radiality cost,” f_1 , is a mandatory cost that must be taken into account, since it expresses the “feasibility” of a decision. If a node of state S_{k+1} is connected to only one node of state S_k , then $f_1 = 0$, otherwise, $f_1 = \infty$. In this way, the formation of loops is avoided.

“Maximum line length cost,” f_2 , prevents the connections of nonneighboring nodes and can be optionally taken into account. If each node of state S_{k+1} is connected with a neighboring node of state S_k , then $f_2 = 0$, else $f_2 = \infty$. The “maximum line length” cost is in accordance with the philosophy of small movements between nodes, which is set as a rule of the problem. A reasonable “maximum line length” value can be $\sqrt{2}D$, where D is the distance between two successive nodes.

“Other costs,” f_3 , can also be introduced to take into account special conditions that may exist for the connection of a node to the neighboring ones, e.g., a high cost underground cable. Obstacles can also be considered by assigning an infinite value to f_3 . The constraint cost is defined as the sum: $f_1 + f_2 + f_3$.

The objective function Z includes the cost of a network section for the whole study period. If a generating unit (e.g., a wind turbine) is connected instead of a load, it can also be handled by the methodology, as explained in the second part of the Appendix.

• *Policy P_k and Policy Cost C_k (Recursive Relationship):* A policy P_k is defined as the set of successive states $(S_k, S_{k+1}, \dots, S_n)$, from stage k to the last stage n , related with the corresponding decisions $X_k, X_{k+1}, \dots, X_n$. The policy cost C_k is the corresponding cost defined as

$$C_k = \sum_{i=k}^n [Z_i(S_i, X_i)] + V_k(P_k) \quad (2)$$

where $Z_i(S_i, X_i)$ is the cost of the decision X_i for the transition from state S_i (of stage i) to state S_{i+1} (of stage $i+1$) and V_k is a virtual cost, attributed to the voltage drop on the network.

Especially for the policies P_0 , additional costs can be added to the corresponding cost C_0 , such as the feeder departure cost.

If a policy P_k is applied, then all states S_k to S_n are known, and therefore a corresponding network is formed and the maximum

imum voltage drop ΔV_k on this network can be calculated. The corresponding cost V_k is 0, if $|\Delta V_k| < \Delta V_{lim}$, and infinite otherwise. ΔV_{lim} is the adopted voltage deviation limit.

If the best policy has been determined for state S_{k+1} , then a decision X_k for the transition from a state S_k to the state S_{k+1} determines a policy P_k with an associated cost C_k . Minimizing C_k for all possible policies P_k results in the optimum policy P_k^* for state S_k with cost C_k^* .

• *Main Procedure:* It can be described by the following steps:

- 0) The GIS grid is determined and the locations of the loads are set.
- 1) Stages 1 to n are determined.
- 2) The states S_k are determined for the last stage n by the procedure outlined below.
- 3) The decisions X_k determined from each state S_k , as it was explained in the decision determination section, and the corresponding costs Z_k are calculated using (1). In stage $k = n$, there is no decision to take from a state S_n , due to the fact that next stage does not exist. Then the cost Z_n for each state can be calculated only for the connections made at state (stage nodes feed all the loads of stage n).
- 4) The policies P_k are determined and the corresponding costs C_k are calculated using (2). The policies P_n for the stage $k = n$ are coincide with the states S_n , so the costs C_n are equal to the corresponding Z_n , if there is no voltage limit violation. The optimal policies P_k^* and the corresponding decisions X_k^* are determined for each state S_k , as described in the policy determination section. For the stage $k = n$, each policy P_n is the optimal for the state S_n , as it is unique.
- 5) The minimum cost criterion described below is applied for the states S_k . The appropriate states S_k and decisions X_k are stored for stage k .
- 6) Steps 2) to 5) are repeated for all stages $k = n-1$ to 0. The procedure stops when the policies P_0 , for stage 0, are determined. Among them, the one presenting the minimum cost is the optimum problem solution (optimum policy P_0^*).

Procedure for the Determination of the States S_k , in a Stage k : This can be done in two ways.

- 1) The states are determined as the feasible ways of supplying all loads L_{kn} of stages k to n , by all nodes M_k of stage k , as explained in the state determination above. However, the number of states obtained in this way is usually very large, so that the method is not practicable.
- 2) Alternatively, the results of stage $k+1$ are used for the determination of the states S_k at stage k , as follows. At a first step, the feasible combinations for the supply of the loads L_k of the stage k by the nodes M_k are determined, resulting in a manageable number of states, since L_k is much smaller than L_{kn} , especially in stages near the substation. These states are called “over-states.” Then, each “over-state” is combined with all states S_{k+1} . In each combination, all possible connection of the “over-state” nodes with the nodes of state S_{k+1} have to be determined, each one representing a state S_k . Hence, each node of

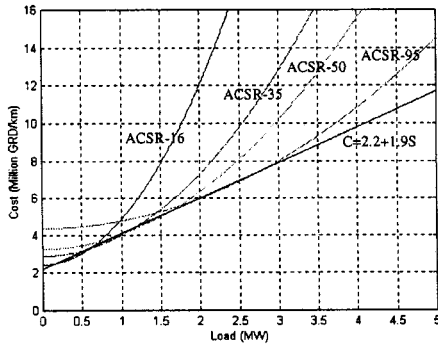


Fig. 5. Linearization of the line cost function.

S_k is assigned the loads of the S_{k+1} node it is connected with.

Minimum Cost Criterion: It is an important artifice in order to reduce the computing effort. From the set of states S_k , which have already been determined for a stage k , a subset can be formed applying a cost criterion on the corresponding policy costs C_k . The best policy among “optimum policies” at that stage will present the minimum cost, $C_k^{\min}$. Defining an arbitrary function associated with stage k , $Cr(k)$, a filtering of the states S_k is made, in order to discard states whose optimum policy presents a cost higher than $C_k^{\lim} = Cr(k) \cdot C_k^{\min}$. The $Cr(k)$ is reduced as the procedure approaches the HV/MV substation, because the effect of the cost Z_k (of going from stage k to $k+1$) on the corresponding policy total cost C_k becomes smaller. A convenient function for the minimum cost criterion is presented in the study cases.

The application of this criterion must be made with caution, in order to avoid the loss of states that might be included in the final optimum solution. A good strategy for its application is to repeat the execution of the algorithm gradually increasing the value of $Cr(k)$ until the final solution remains unchanged in two or three successive iterations.

III. MODEL VALIDATION

In order to validate the model proposed in this paper, its results for the simple problem shown in Fig. 1 are compared with those obtained using the mixed integer programming (MIP) technique, implemented in the GAMS software, [17]. In this technique, a linearized cost function represents the line cost. The cost function for the transportation of a power S (MW) is $C = 2.2 + 1.9 \cdot S$ [million GRD]. This function fits well to the optimum cross section selection points for four “normalized” conductor types: ACSR-95, 50, 35, and 16. The reliability cost is ignored.

The cost of each line has been calculated using (A3) of the Appendix for a study period of $N = 10$ years, load growth rate $\tau = 4\%$, load factor $LD = 0.4$, discount rate $i = 6\%$, and nominal voltage $V_0 = 20$ kV.

In Fig. 1, the load and substation positions and the initial maximum demand of each load are indicated. It was assumed that all loads appear in year 0. The GIS grid size is $2 \text{ km} \times 2 \text{ km}$ per “pixel.” The optimum routing of one or more 20-kV radial

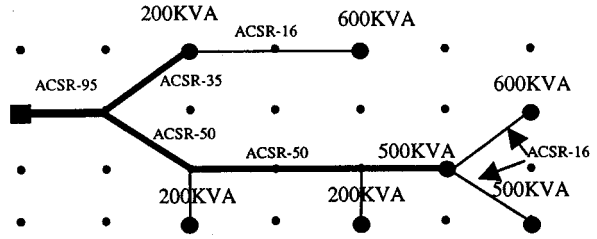


Fig. 6. Network configuration proposed from both methods.

feeders is sought, along with the selection of the optimum cross sections among the four types of Fig. 5. It is considered that all pixels are candidate for the passage of the lines. The solution to the problem is shown on Fig. 6 and it is identical for both algorithms.

The computation time using GAMs was 75 min, while using the proposed algorithm was only 23 s. The cost criterion, as a function of stage k , $Cr(k)$ was taken

$$Cr(k) = Cr_0 + e^{ak}$$

where a and Cr_0 are constant values. The values selected are $a = 0.0676$ and $Cr_0 = 0.1$, so that, the cost criterion value is 1.6 at the last stage ($k = 6$) and 0.1 at stage $k = 0$.

IV. STUDY CASE

In Fig. 7(a), a realistic study case routing problem is presented. 25 loads, dispersed in a region of $30 \times 20 = 600 \text{ km}^2$, have to be fed by an HV/MV substation. A suitable 20-kV network is sought presenting a minimum total cost for a 5-year period. A maximum voltage drop limit is set at 8%. The initial load values and the starting years are indicated on Fig. 7(a). The load growth rate is $\tau = 8\%$ for all loads and the discount rate $i = 6\%$.

Two cases are considered. In the first, existing obstacles, shown in Fig. 7(a), are taken into account, while in the second, they are ignored. The cost criterion is the same as in Section III.

The results obtained are shown in Fig. 7(b) and (c), respectively. It is clear that the network configurations obtained in both cases are very close to each other, while the obstacles are properly avoided in Fig. 7(b), increasing slightly the total cost. This problem was also attempted to be solved using GAMs, but the enormous time required did not permit its solution.

V. CONCLUSION

The optimal routing of distribution networks was addressed in this paper, using a dynamic programming technique and GIS facilities. The advantage of the developed algorithm is that the total cost (investment, transmission losses and reliability) for a multiyear study period is optimized, taking into account all constraints, technical (voltage drop, thermal limits) and physical (obstacles, high cost passages, etc.), as well as existing line sections. Dispersed generation can be also taken into account. Consequently, the developed algorithm can be effectively applied in realistic problems, where all the above mentioned factors exist. In addition, it could be incorporated in a general optimum planning distribution network software.

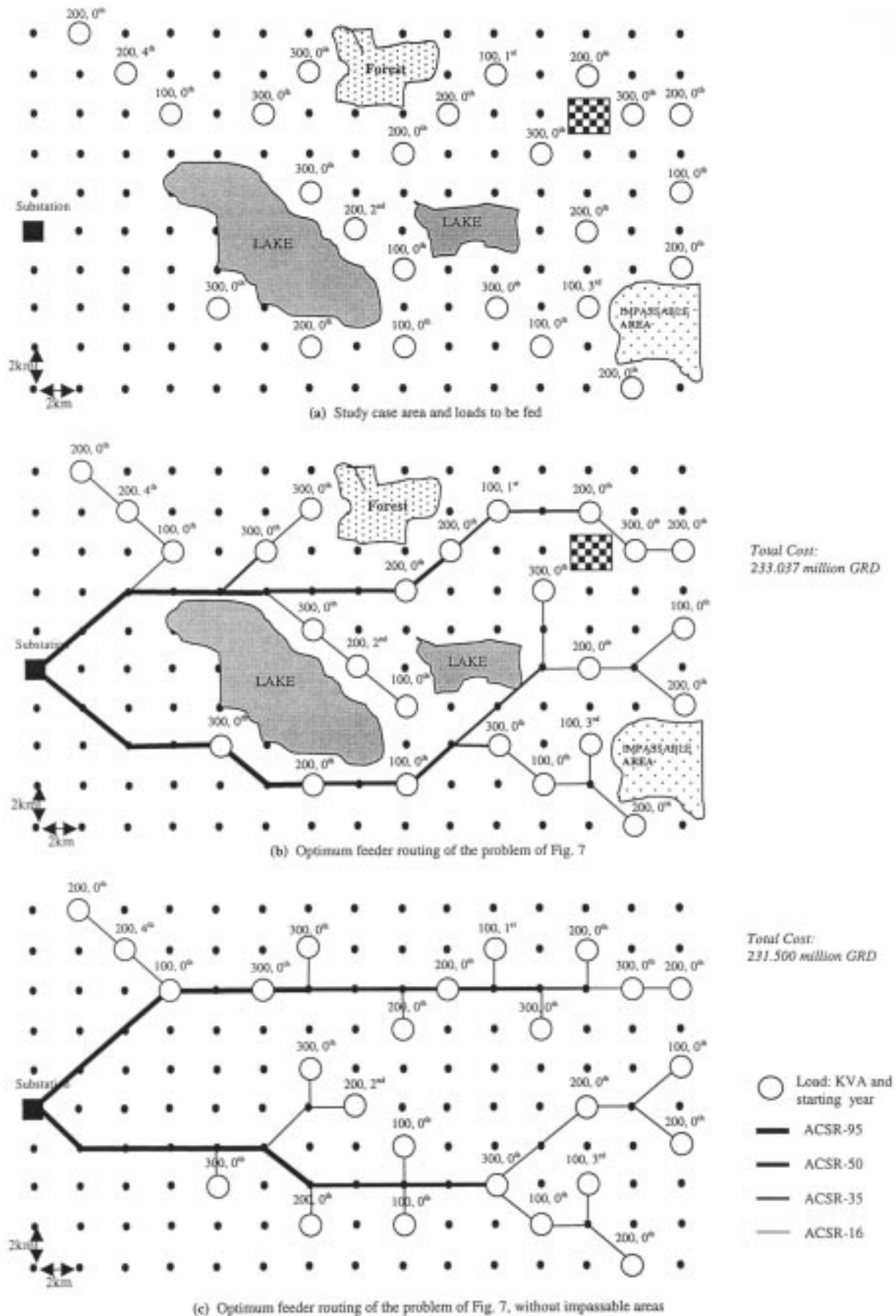


Fig. 7. Study case.

APPENDIX

SELECTION OF OPTIMUM CONDUCTOR CROSS SECTION

A. Loads

The optimum cross section q per unit length (km) of line is calculated as follows.

The total cost K_{tot} of a line section with an expected life period of N_{life} years and a utilization period of N years is considered as the sum of the investment cost K_c and the cost of losses K_{loss} . The reliability cost is ignored, as it can be assumed to be independent from the cross section q . Hence

$$K_c = (K_1 + K_2 q) A(i, N_{\text{life}}) N = K_1^* + K_2^* \cdot q \quad (\text{A1})$$

where

$$A(i, N_{\text{life}}) = i \cdot (1 + i)^{N_{\text{life}}} / [(1 + i)^{N_{\text{life}}} - 1]$$

K_1 , K_2 are constant values related to line investment cost and

$$K_{\text{loss}} = (a + b \cdot LF \cdot 8760) \cdot \frac{\rho}{q} \frac{S_0^2}{V_0^2} \sum_{n=1}^N \frac{(1 + \tau)^{2n}}{(1 + i)^n} = K_3 \frac{S_0^2}{q} \quad (\text{A2})$$

where $LF = 0.3LD + 0.7LD^2$ is the loss factor as a function of the load factor (LD); a (GRD/kW) and b (GRD/kWh) are cost of power and energy losses; ρ is the resistivity of the conductor; τ is the annual growth rate of the load; S_0 is the initial load (in year 0); and V_0 is nominal voltage.

Consequently

$$K_{\text{tot}} = K_c + K_{\text{loss}} = K_1^* + K_2^* \cdot q + \frac{K_3 \cdot S_0}{q} \quad (\text{A3})$$

Setting the first derivative with respect to q equal to zero leads to

$$q_{\text{opt}} = S_0 \cdot \sqrt{\frac{K_3}{K_2^*}} \quad (\text{A4})$$

Generally speaking, the load and the loss factor for a group of loads can be easily calculated. Consequently, the optimum cross-section size of the feeding line can be selected by this formula.

B. Dispersed Generation

Dispersed generation can be also included in the algorithm. For instance, if a wind farm is connected, the Weibull distribution of wind and the wind turbine power versus wind speed curve can be used for calculating the expected wind power $S_{w,k}$ for a time period Δt_k . The corresponding line loss cost for a line with resistance R_0 is calculated by the following formula for each year n :

$$K_{w,\text{loss},n} = \frac{R_0}{V_0^2} \sum_{k=1}^M S_{w,k}^2 (a + b \cdot \Delta t_k) \quad (\text{A5})$$

where: M is the number of wind speed bins of the power–wind speed curve.

Following the same procedure as for the loads, it is obtained

$$q_{\text{opt}} = \sqrt{\frac{K_3^*}{K_2^*}} \quad (\text{A6})$$

where K_3^* is given by the formula

$$K_3^* = \frac{\rho}{V_0^2} \cdot \sum_{n=1}^N \frac{K_{w,\text{loss},n}}{(1 + i)^n} \quad (\text{A7})$$

REFERENCES

- [1] H. L. Willis, *Power Distribution Planning Reference Book*. New York: Marcel Dekker, 1997.
- [2] R. N. Adams and M. A. Laughton, "Optimal planning of power networks using mixed-integer programming," *Proc. Inst. Elec. Eng.*, vol. 121, no. 2, Feb. 1974.
- [3] E. Masud, "An interactive procedure for sizing and timing distribution substations using optimal techniques," in *Proc. IEEE PES Winter Meet.*, New York, Jan. 27, 1974.
- [4] D. M. Crawford and S. B. Holt, "A mathematical optimization technique for locating and sizing distribution substations, and deriving their optimal service areas," *IEEE Trans. Power App. Syst.*, vol. PAS-94, Mar./Apr. 1975.
- [5] D. L. Wall, G. L. Thompson, and J. E. D. Northcote-Green, "An optimization model for planning radial distribution networks," *IEEE Trans. Power App. Syst.*, vol. PAS-98, May–June 1979.
- [6] D. I. Sun, D. R. Ferris, P. J. Cote, R. R. Shoults, and M. S. Chen, "Optimal distribution substation and primary feeder planning via the fixed charge network formulation," *IEEE Trans. Power App. Syst.*, vol. PAS-101, Mar. 1982.
- [7] M. A. El-Kady, "Computer-aided planning for distribution substation and primary feeders," *IEEE Trans. Power App. Syst.*, vol. PAS-103, June 1984.
- [8] T. H. Fawzi, K. F. Ali, and S. M. El-Sobki, "A new planning model for distribution systems," *IEEE Trans. Power App. Syst.*, vol. PAS-102, Sept. 1983.
- [9] T. Gonen and B. L. Foote, "Distribution-system planning using mixed-integer programming," *Proc. Inst. Elec. Eng.*, pt. C, vol. 128, no. 2, Mar. 1981.
- [10] T. Gonen and I. J. Ramirez-Rosado, "Optimal expansion on power distribution systems—Part I: Basic modeling," *Int. J. Energy Syst.*, vol. 10, no. 2, 1990.
- [11] —, "Optimal expansion on power distribution systems—Part II: Voltage drop and radiality constraints," *Int. J. Energy Syst.*, vol. 10, no. 2, 1990.
- [12] M. Ponnavaikko, K. S. P. Rao, and S. S. Venkata, "Distribution systems planning through a quadratic mixed integer programming approach," *IEEE Trans. Power Delivery*, vol. PWRD-2, Oct. 1987.
- [13] S. Jonnavithula and R. Billinton, "Minimum cost analysis of feeder routing in distribution system planning," *IEEE Trans. Power Delivery*, vol. 11, Oct. 1996.
- [14] G. J. Peponis and M. P. Papadopoulos, "New dynamic, branch exchange method for optimal distribution system planning," *Proc. Inst. Elec. Eng.—Gener., Transm. Distrib.*, vol. 144, no. 3, May 1997.
- [15] N. G. Boulaxis, M. P. Papadopoulos, and G. Glinou, "Planning of distribution networks in high wind penetration areas using GIS facilities," in *Proc. CIRED '99*, Nice, France, June 1999.
- [16] F. S. Hillier and G. J. Liberman, *Introduction to Operation Research*. New York: McGraw-Hill, 1989, pp. 266–270.
- [17] "GAMS, A User's Guide," GAMS Development Corp., Dec. 1998.

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